Question 1 (1 point)

What is the primary difference between analog communication and digital communication?

A. Digital communication uses wireless channels while analog communication uses wired channels.

B. Digital communication uses a digital source while analog communication uses a continuous-time source.

C. Digital communication does not use analog hardware.

D. Digital communication is always more secure.

Question 2 (1 point)

In a digital communication system, what is the primary purpose of a source encoder?

A. Correct channel errors

B. Compress the information source

C. Modulate the carrier

D. Encrypt data

Question 3 (1 point)

Which block is responsible for adding redundancy to combat transmission errors?

A. Source Coding

B. Encryption

C. Channel Coding

D. Modulation

Question 4 (1 point)

A channel code with rate $1/2$ means:

A. Half of the transmitted bits are discarded.

B. One information bit produces two coded bits.

C. Two coded bits produce one information bit at the receiver only.

D. The bit error rate is reduced by 50%.

Question 5 (1 point)

What is the primary function of a modulator?

A. Detect symbols

B. Compress data

C. Map bits to analog waveforms

D. Remove channel impairments

Question 6 (1 point)

What is the purpose of a demodulator?

A. Recover transmitted information from received signals

B. Amplify the transmitted signal

C. Generate encryption keys

D. Increase transmission power

Question 7 (1 point)

Which of the following is a common wireless channel impairment?

A. Path loss

B. Multipath propagation

C. Thermal noise

D. All of the above

Question 8 (1 point)

Why are software-defined radios (SDRs) attractive?

A. They eliminate the need for antennas.

B. They can be reconfigured through software.

C. They operate without analog circuitry.

D. They do not require modulation

Question 9 (1 point)

True or False:

Digital communication systems ultimately transmit electromagnetic waves through the channel.

_____

Question 10 (1 point)

Place the following blocks in the correct transmitter order:

- Channel Coding
- Source Coding
- Modulation

Answer:

_____ → _____ → _____

Answer Key

1. **B**
2. **B**
3. **C**
4. **B**
5. **C**
6. **A**
7. **D**
8. **B**
9. **True**
10. **Source Coding → Channel Coding → Modulation**
